DATA SCIENCE & ANALYTICS

MODULE 03 SESSION 15

Forecasting, explaining and serving predictions

What happens next, why did the model say that, and how does a report get it?

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Mon 26 Oct 2026

BY THE END OF TODAY, YOU CAN

Class objectives

01 Forecast a series with ARIMA_PLUS; read its interval

02 Split a time series by date, not at random

03 Explain one prediction and the whole model

04 Stop a preprocessing leak with TRANSFORM

05 Save predictions as ml.* tables for Power BI

EVERYONE DO

Step 1.1

Warm-up

Five questions on D14.

Suggested time · 10 minutes

PART 01

A forecast, before the theory

Load twelve years of monthly passengers, train a prepared ARIMA_PLUS model, read its forecast.

INSTRUCTOR DO

Step 2.1

Today’s question

An airline planner, and two questions from D14.

Suggested time · 4 minutes

TODAY · ONE STORY, THREE QUESTIONS

Three questions for today

Part	Question	Data
1–3	How many passengers will fly each month next year?	Air Passengers: 144 months, 1949–1960
4	Why did D14’s trees charge person E about 43,500 dollars?	D14’s insurance model
5	How does a Power BI report get these numbers?	two ml.* tables, read on D22

An airline planner needs the first answer to plan aircraft and crews. The numbers are thousands of passengers a month.

EVERYONE DO

Step 2.2

Load the series and look at it

Upload air_passengers.csv as ml.air_passengers, count the months, see the climb.

Suggested time · 12 minutes

labs/D15/01-Evr-Monthly-Series/README.md

STEP 1 · ONE ROW IS ONE MONTH

How many months, from when to when?

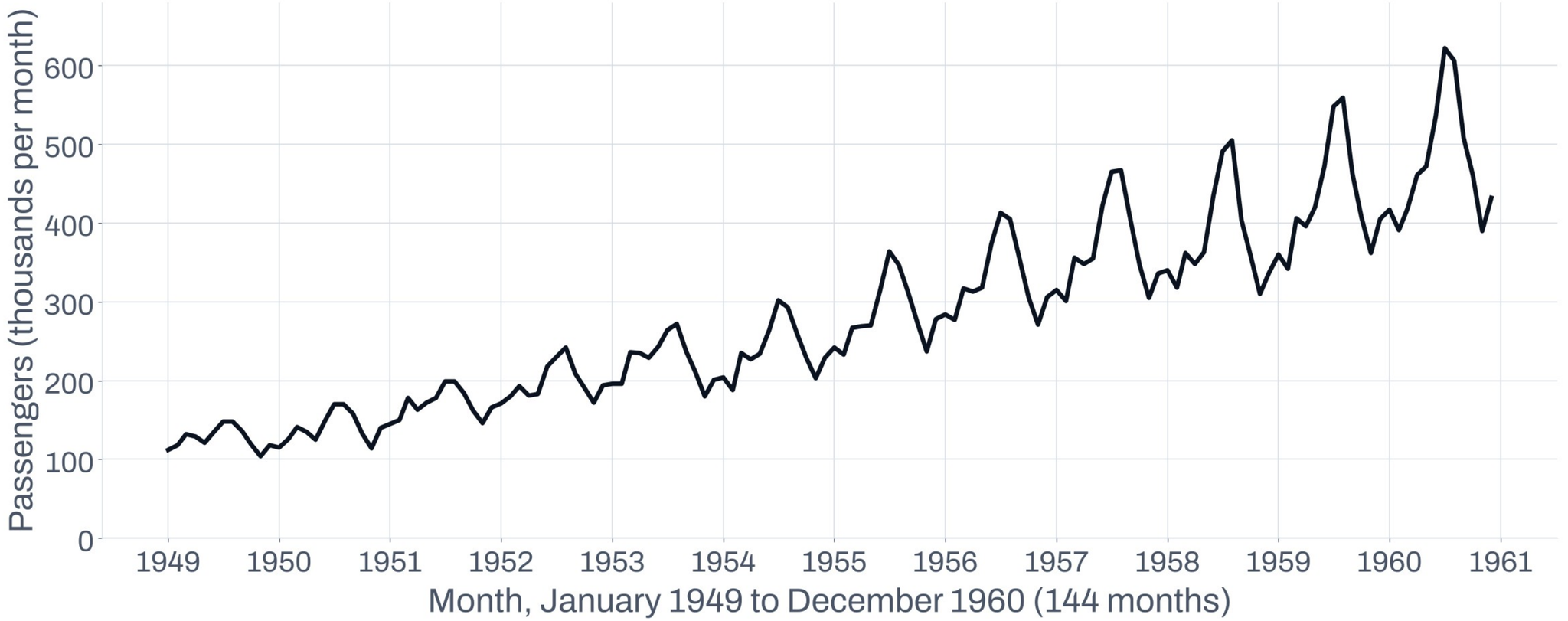
COUNT, MIN and MAX on the DATE column: none missing, none extra.

```
SELECT COUNT(*) AS months,
       MIN(month) AS first_month,
       MAX(month) AS last_month
FROM ml.air_passengers;
```

→ 144 · 1949-01-01 · 1960-12-01

ALL 144 MONTHS · THOUSANDS OF PASSENGERS

A climb, and a summer peak that grows



Step 3 adds up each year: 1,520 in 1949, 5,714 in 1960, every year above the one before.

EVERYONE DO

Step 2.3

A prepared forecast

Train ARIMA_PLUS on the first ten years, then ask it for twelve months.

Suggested time · 14 minutes

labs/D15/02-Evr-First-Forecast/README.md

ARIMA_PLUS on the 120 training months

`MODEL_TYPE = 'ARIMA_PLUS'`: BigQuery's forecasting model. ml.air_training holds the months before 1959.

```
CREATE OR REPLACE MODEL ml.air_arima
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_training;
```

STEP 2 · THREE OPTIONS

Time, the value, and how far ahead

TIME_SERIES_TIMESTAMP_COL, **TIME_SERIES_DATA_COL**, **HORIZON**: no label column. The series is its own label.

```
CREATE OR REPLACE MODEL ml.air_arima
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_training;
```

HORIZON = 12 asks for twelve months ahead. Google’s default is 1,000.

STEP 3 · TWELVE MONTHS AHEAD

Twelve months, each with a range

`STRUCT(12 AS horizon, 0.95 AS confidence_level)`: twelve months, at 95%.

```
SELECT forecast_timestamp, forecast_value,
       prediction_interval_lower_bound,
       prediction_interval_upper_bound
FROM ML.FORECAST(MODEL ml.air_arima,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level));
```

```
→ 12 rows · Jan 1959 about 353 · Aug about 523
```

The range widens further ahead: January about 341 to 364.

INSTRUCTOR DO

Step 2.4

How would we know it is any good?

It climbs and peaks in summer. Which months do we hold aside, and what simple guess must it beat?

Suggested time · 5 minutes

PART 02

What a forecast is

Trend and season, split by date, two simple forecasts, and what ARIMA_PLUS does.

INSTRUCTOR DO

Step 3.1

A time series: trend, season, noise

Rows in an order, and three parts inside them.

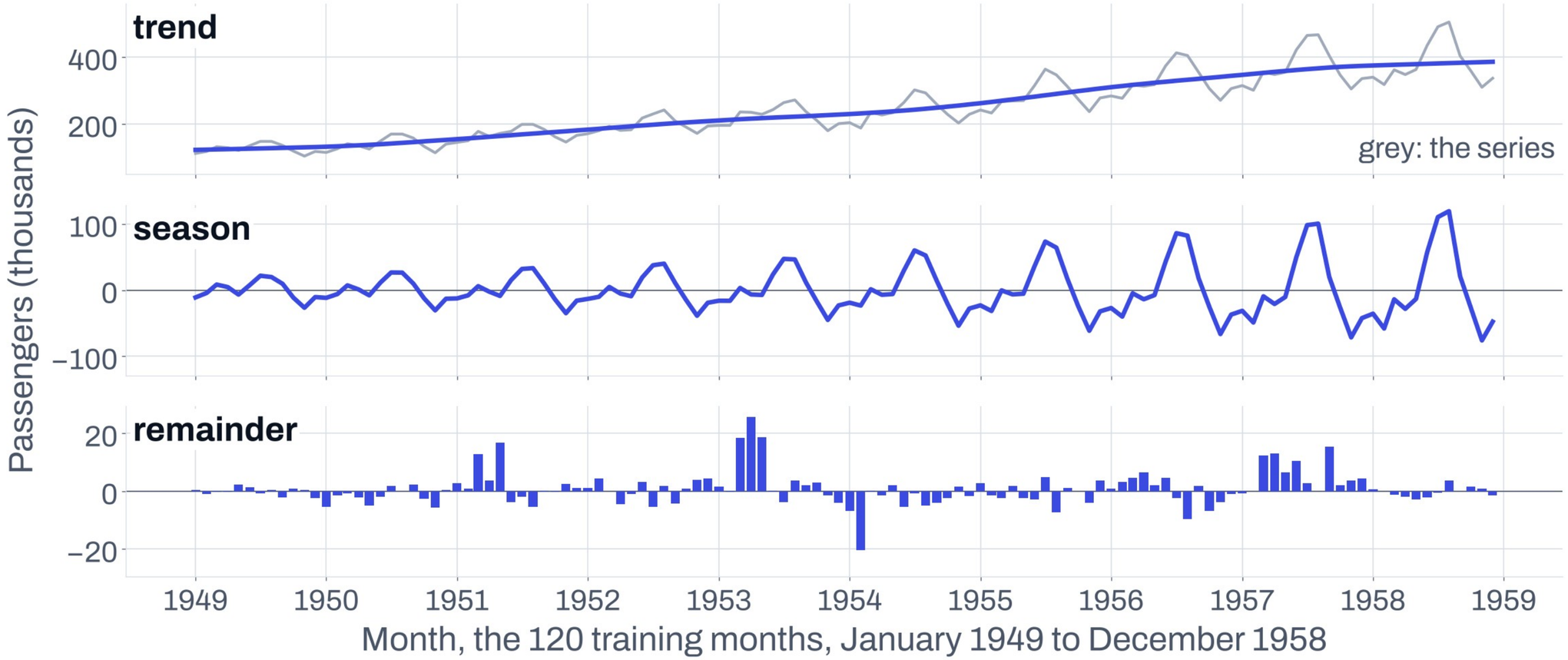
Suggested time · 10 minutes

DEFINITION

A time series is a set of values measured at regular time points, in order. One row is one month; shuffle the months and the pattern is gone.

THE 120 TRAINING MONTHS, IN THREE PARTS

Trend + season + remainder = the series



Trend: the long climb. Season: a peak in July and August, a low in November. Remainder: what is left.

INSTRUCTOR DO

Step 3.2

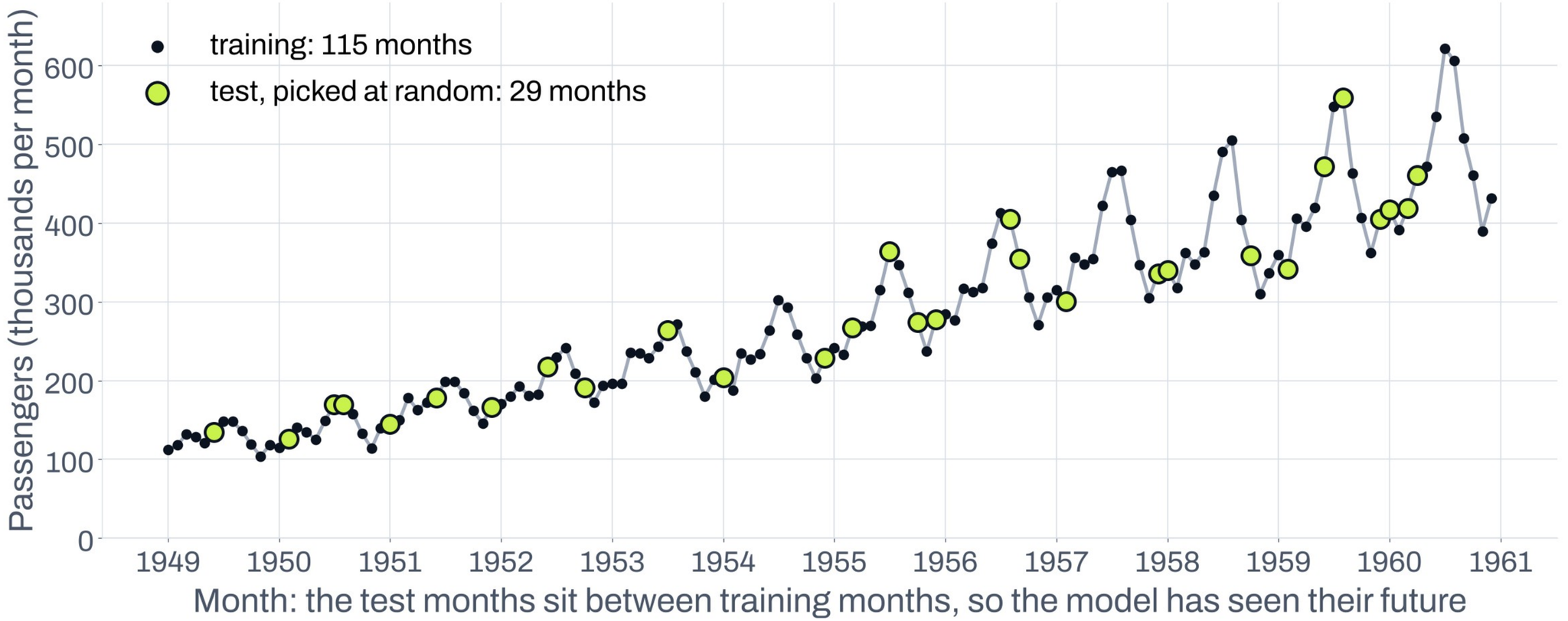
Split by date, never at random

Which months to hold aside, and how far ahead to forecast.

Suggested time · 8 minutes

D11’S RANDOM SPLIT, ON A TIME SERIES

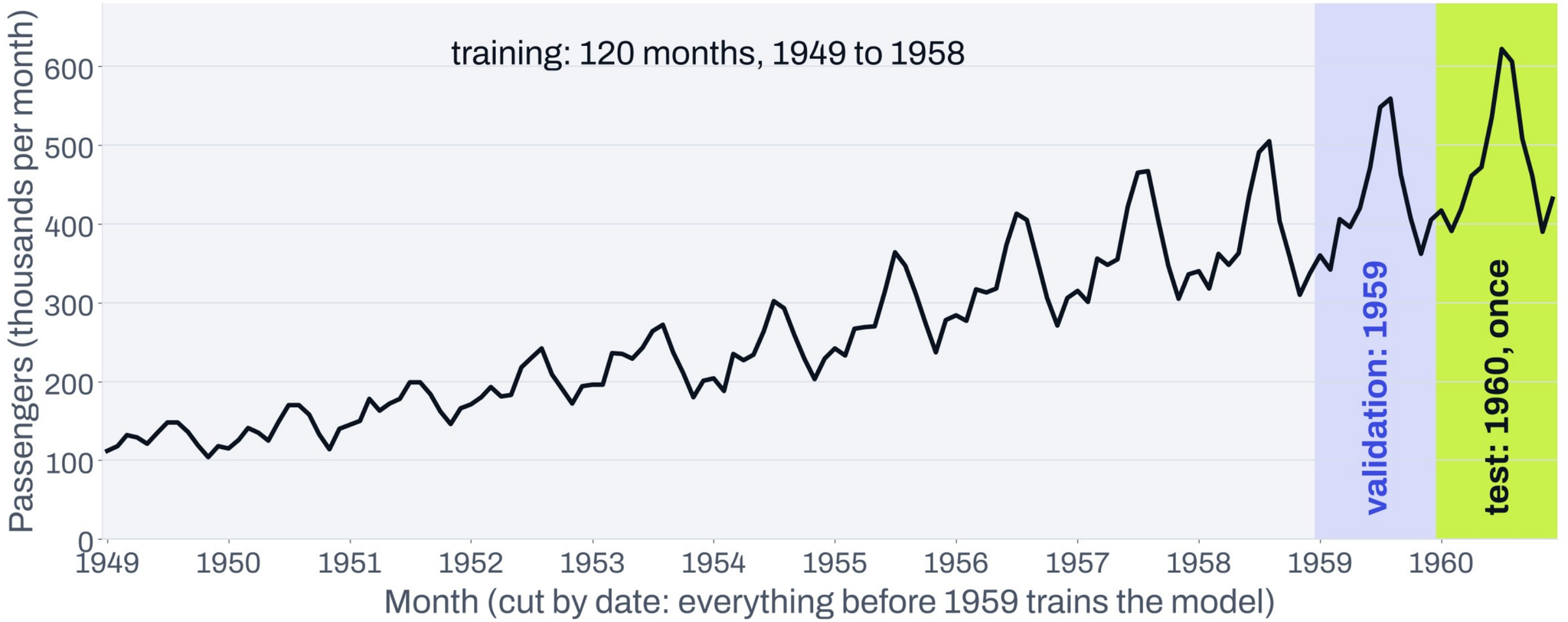
A random split lets the model see the future



Every held-out month has training months on both sides. A real forecast never knows next month.

TRAINING · VALIDATION · TEST, BY DATE

Before 1959 trains; 1959 compares; 1960 tests once



Split by date: months before a cutoff train the model; the months after it are scored, in order.

One cutoff, several months ahead

The horizon: how far ahead we forecast. Ours is 12 months.

To Dec 1958	Jan 1959	Feb 1959	Dec 1959
120 months	h=1	h=2	h=12

No 1959 month is known when the forecast is made.

The naive forecast repeats the last month we know for every month ahead: December 1958, 337 thousand, held flat across 1959.

The seasonal naive repeats the same month one year earlier: January 1959 gets January 1958 (340), August 1959 gets August 1958 (505).

STUDENT DO

Split by date and two simple forecasts

Make the validation and test views, then score the naive and the seasonal naive on 1959 in plain SQL.

Suggested time

12 minutes

labs/D15/03-Stu-Split-By-Date/README.md

STUDENT DO

Two views, two forecasts, one score each

INSTRUCTIONS	Create ml.air_validation (1959) and ml.air_test (1960) by date. Score the naive forecast (337 every month) and the seasonal naive on 1959: the average miss, signs dropped (MAE).
HINT	Which training month should January 1959 be compared with?
BONUS	Score the training mean and median too. Which does worst, and why?

Full steps and Check yourself: labs/D15/03-Stu-Split-By-Date/README.md

REVIEW

Time's up.
Let's review.

Split by date and two simple forecasts

`Solved: labs/D15/03-Stu-Split-By-Date/Solved/split_by_date.sql`

Each 1959 month meets the same month of 1958

`DATE_SUB(v.month, INTERVAL 1 YEAR)`: January 1959 joins January 1958.

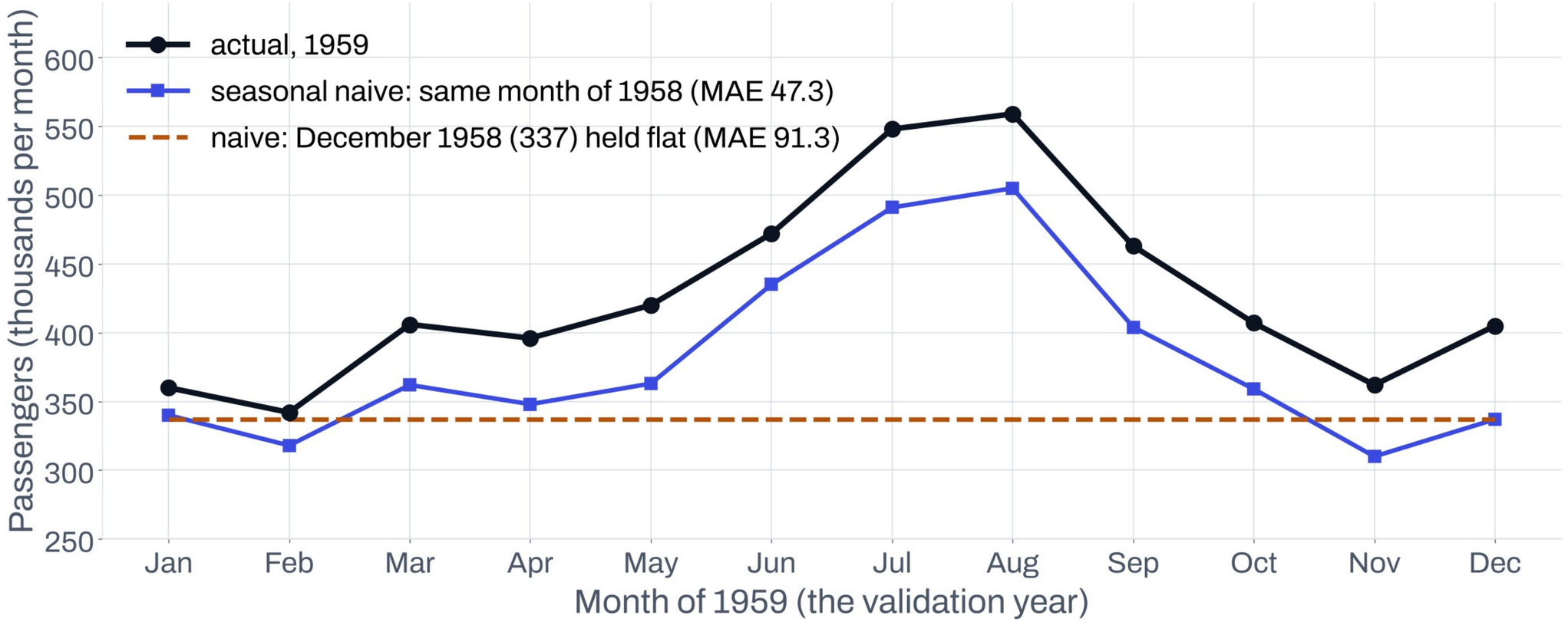
```
SELECT ROUND(AVG(ABS(v.passengers - t.passengers)), 1)
         AS mae_seasonal_naive,
       ROUND(100 * AVG(ABS(v.passengers - t.passengers)
              / v.passengers), 1) AS mape_seasonal_naive
FROM ml.air_validation AS v
JOIN ml.air_training AS t
  ON t.month = DATE_SUB(v.month, INTERVAL 1 YEAR);


                    → 47.3 · MAPE 11.1 · the naive, 337 held flat: 91.3
```

Joined on v.month = t.month, it returns no rows.

VALIDATION · 1959 · MAE IN THOUSANDS OF PASSENGERS A MONTH

Last year’s month halves the naive’s miss



Training mean: 182.4. Median: 194.8, worse. Both guess one flat number for a series that grows.

INSTRUCTOR DO

Step 3.5

What ARIMA_PLUS does

Clean the series, find the season, model the trend, add the parts back.

Suggested time · 10 minutes

ARIMA_PLUS · GOOGLE’S PAGE, IN ORDER

Four steps, in words

Step	What BigQuery does	On Air Passengers
1 clean	fills gaps, averages duplicates, cleans spikes and dips, adjusts step changes	no gaps; holiday effects apply only to daily or weekly series
2 season	splits off the yearly pattern and carries it forward	the July and August peak
3 trend	fits an ARIMA to what is left, choosing its order itself	the climb
4 add	forecasts each part and adds them up	trend + season, 12 months ahead

The range widens with the horizon: each month further out stacks up more uncertainty.

STEP 3 · ARIMA(p, d, q), ONE LINE EACH

The trend model in plain words

Part	In words	A Python check chose
AR, p	next month leans on the last few months’ values	p = 0
I, d	work with month-to-month changes, not levels	d = 1
MA, q	next month leans on the last few months’ surprises	q = 1
drift	a steady climb added to those changes	yes

AUTO_ARIMA tries candidates and keeps the lowest AIC; ML.ARIMA_EVALUATE shows BigQuery’s pick. The season is added back as a fixed amount. What if it grows?

PART 03

Judge the forecast

Score 1959 against the simple forecasts, read the interval, then test 1960 once.

EVERYONE DO

Step 4.1

Score the forecast on 1959

Join the forecast to the actual 1959 months and score it like the simple forecasts.

Suggested time · 15 minutes

labs/D15/04-Evr-Score-The-Forecast/README.md

STEP 2 · THE SCORE, IN THE SAME UNITS

The forecast against the 1959 months

`DATE(f.forecast_timestamp)`: the forecast's TIMESTAMP becomes a month that joins the DATE column.

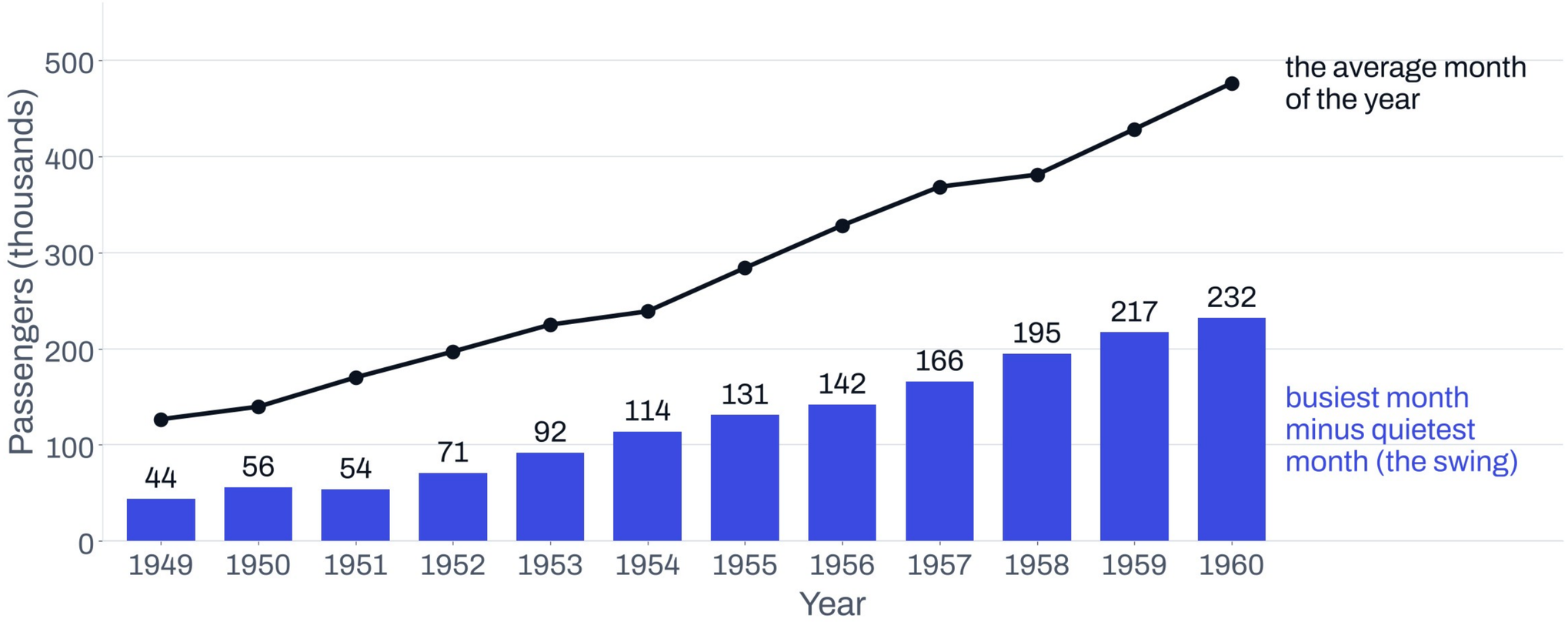
```
SELECT ROUND(AVG(ABS(v.passengers - f.forecast_value)), 1) AS mae,
       ROUND(100 * AVG(ABS(v.passengers - f.forecast_value)
             / v.passengers), 1) AS mape_pct
FROM ML.FORECAST(MODEL ml.air_arima,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level)) AS f
JOIN ml.air_validation AS v
  ON DATE(f.forecast_timestamp) = v.month;
```

→ MAE about 27 · MAPE about 6.2%

The seasonal naive missed by 47.3: the model beats it clearly.

BUSIEST MONTH MINUS QUIETEST MONTH, YEAR BY YEAR

The summer swing grows as the airline grows



From 44 in 1949 to 232 in 1960. A season added as a fixed amount comes in low in the 1959 summer.

INSTRUCTOR DO

Step 4.2

Reading the interval

What the range means, why it widens, and which bound a planner uses.

Suggested time · 8 minutes

DEFINITION

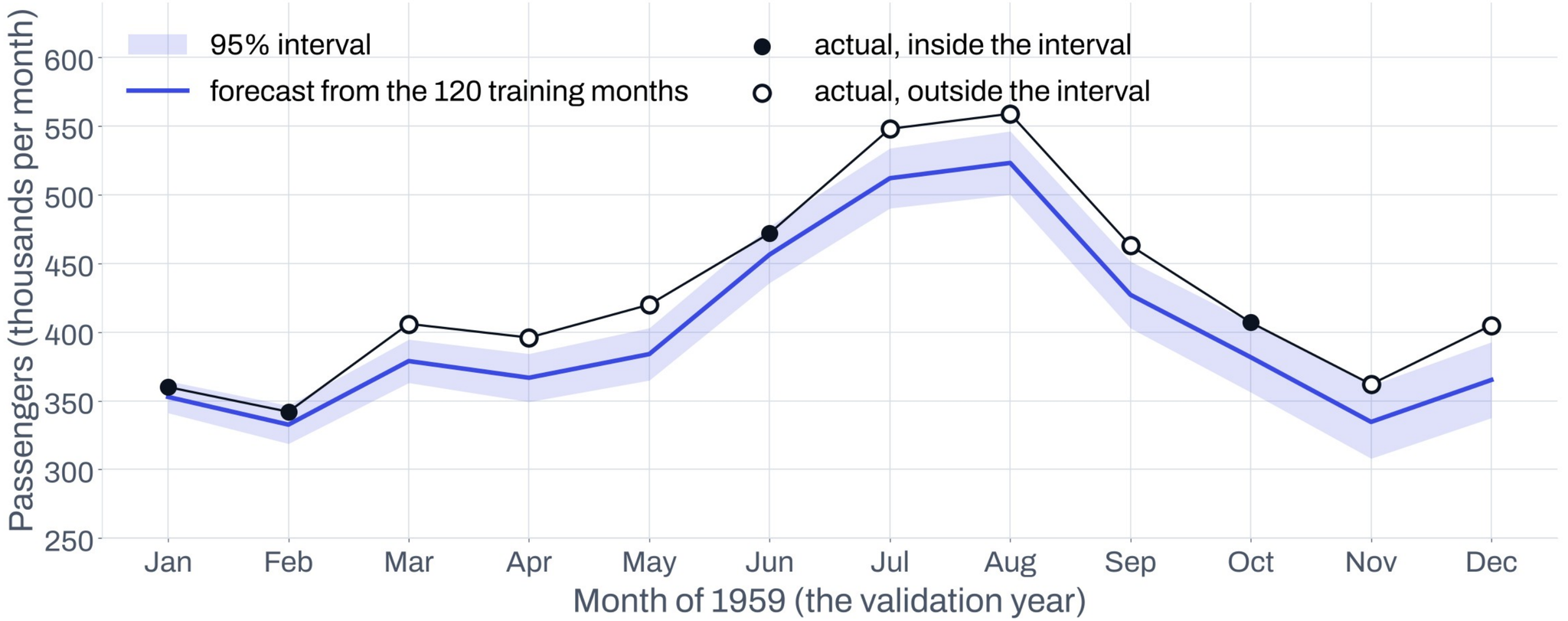
Prediction interval

A 95% prediction interval is the range that should hold one future month’s actual value 95 times in 100.

The confidence_interval columns are narrower: they cover the average, not one month.

VALIDATION · 1959 · THE FORECAST AND ITS 95% INTERVAL

A 95% interval that held about 4 of the 12 months



From a Python stand-in that follows ARIMA_PLUS's steps; BigQuery's own interval may hold more.

EVERYONE DO

Step 4.3

Months that look unusual

Every training month split into its parts. What trend and season leave over is where odd months show.

Suggested time · 8 minutes

labs/D15/04-Evr-Score-The-Forecast/README.md

STEP 4 · ML.EXPLAIN_FORECAST

Months the parts do not explain

ML.EXPLAIN_FORECAST: every training month split into its parts.

```
SELECT DATE(time_series_timestamp) AS month_history, time_series_data,
       ROUND(trend, 1) AS trend, ROUND(seasonal_period_yearly, 1) AS season,
       spikes_and_dips, ROUND(residual, 1) AS residual
FROM ML.EXPLAIN_FORECAST(MODEL ml.air_arima,
                         STRUCT(12 AS horizon, 0.95 AS confidence_level))
WHERE time_series_type = 'history'
ORDER BY ABS(IFNULL(spikes_and_dips, 0) + IFNULL(residual, 0)) DESC
LIMIT 5;
```

→ Apr 1953 +26 · Feb 1954 −20 · May 1953 +19 (about)

Thousands above or below trend + season: a question, not a deletion.

PARTNER DO

Would you plan on this forecast?

Read the 1959 scores and the interval chart. Three sentences for the planner: which forecast, which bound, one risk.

Suggested time

12 minutes

labs/D15/05-Par-Plan-On-The-Forecast/README.md

PARTNER DO

Three sentences for the planner

INSTRUCTIONS	With your partner, use the 1959 scores and the interval chart. Write three sentences: which forecast to plan 1960 on, which bound to staff aircraft and crews to, and one risk. Post them.
HINT	In which months did the model miss most, and in which direction?
BONUS	Why staff to a bound rather than to the forecast itself?

Full steps and Check yourself: labs/D15/05-Par-Plan-On-The-Forecast/README.md

EVERYONE DO

Step 4.5

The test year, once

Retrain on every month before 1960, forecast 1960, and score it one time.

Suggested time · 8 minutes

labs/D15/06-Evr-Test-Year-Once/README.md

STEP 1 · THE CHOICE IS MADE

Retrain on 132 months, then score 1960

`WHERE month < '1960-01-01'`: training and validation together, 1949 to 1959.

```
CREATE OR REPLACE MODEL ml.air_arima_final
OPTIONS (MODEL_TYPE = 'ARIMA_PLUS',
         TIME_SERIES_TIMESTAMP_COL = 'month',
         TIME_SERIES_DATA_COL = 'passengers',
         HORIZON = 12) AS
SELECT month, passengers FROM ml.air_passengers
WHERE month < '1960-01-01';

                    → 1960: about 18 · seasonal naive from 1959: 47.8
```

Run it once: a second try turns 1960 into another validation year.

REVIEW

Time's up.
Let's review.

Judge the forecast

`Solved: labs/D15/05-Par-Plan-On-The-Forecast/Solved/plan_on_the_forecast.md`

A SAMPLE ANSWER FOR THE PLANNER

Three sentences

#	Sentence
1	Plan 1960 on ARIMA_PLUS’s forecast: off by about 27 thousand passengers a month on 1959, against 47.3 for last year’s same month.
2	Staff aircraft and crews to the upper bound in summer: running short costs more than a few empty seats.
3	Risk: the season grows with the airline, so the summer peaks come in above the forecast.

THE SCOREBOARD · MAE, THOUSANDS OF PASSENGERS A MONTH

ARIMA_PLUS beats every simple forecast

Forecast	1959 (validation)	1960 (test, once)
training mean	182.4	–
training median	194.8	–
naive: last month held flat	91.3	76.0
seasonal naive: same month a year earlier	47.3	47.8
ARIMA_PLUS	**about 27**	**about 18**

Lower is better. On 1960 the simple forecasts start from 1959. The ARIMA_PLUS figures come from a stand-in; yours may differ a little.

Add a fixed amount, or a share of the level?

Level	Season added as +10	Season as a share, ×1.1
100	110	110
200	210	220
400	410	**440**

ARIMA_PLUS adds. Forecast the logarithm of passengers and convert back, and a share becomes an amount: the stand-in’s 1959 miss drops from about 27 to about 17.

Break

30 minutes

After the break: why did D14’s trees charge person E so much?

PART 04

Why did the model say that?

One prediction split into pushes, then the model overall. D14’s insurer and its boosted trees.

INSTRUCTOR DO

Step 5.1

One prediction, many reasons

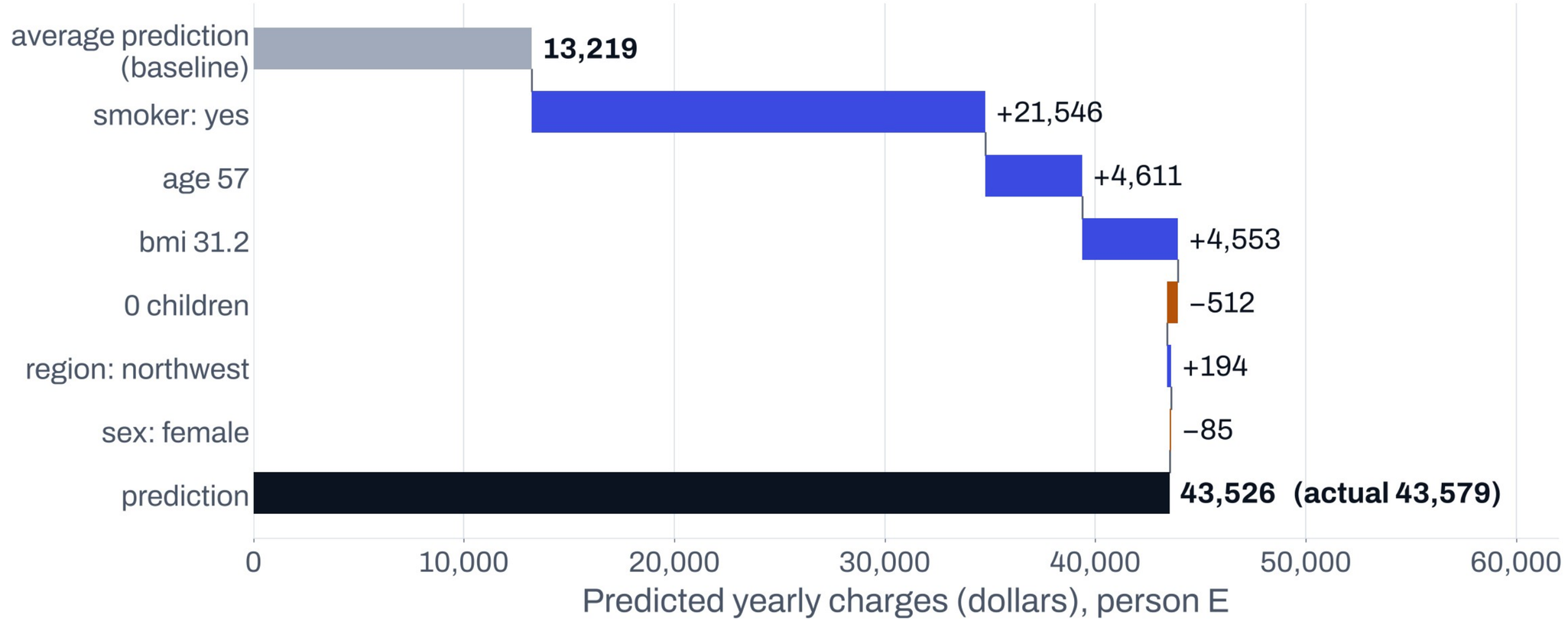
Person E: 57, female, smoker, BMI 31.2. Predicted about 43,526 dollars; actual 43,579.

Suggested time · 6 minutes

An attribution is how far one feature’s value pushed a prediction up or down from the baseline, the model’s average prediction over its training rows.

PERSON E · D14’S BOOSTED TREES · DOLLARS A YEAR

Baseline + pushes = the prediction



About 13,219 + 21,546 + 4,611 + 4,553 − 512 + 194 − 85 = 43,526. Smoking is about half of it.

EVERYONE DO

Step 5.2

Explain five people

ML.EXPLAIN_PREDICT on D14’s boosted model, for people A to E.

Suggested time · 14 minutes

labs/D15/07-Evr-Explain-Five-People/README.md

STEP 1 · ONE PREDICTION, SPLIT INTO PUSHES

ML.EXPLAIN_PREDICT: ML.PREDICT plus the reasons

top_k_features: pushes per row, largest first; 6 shows all six.

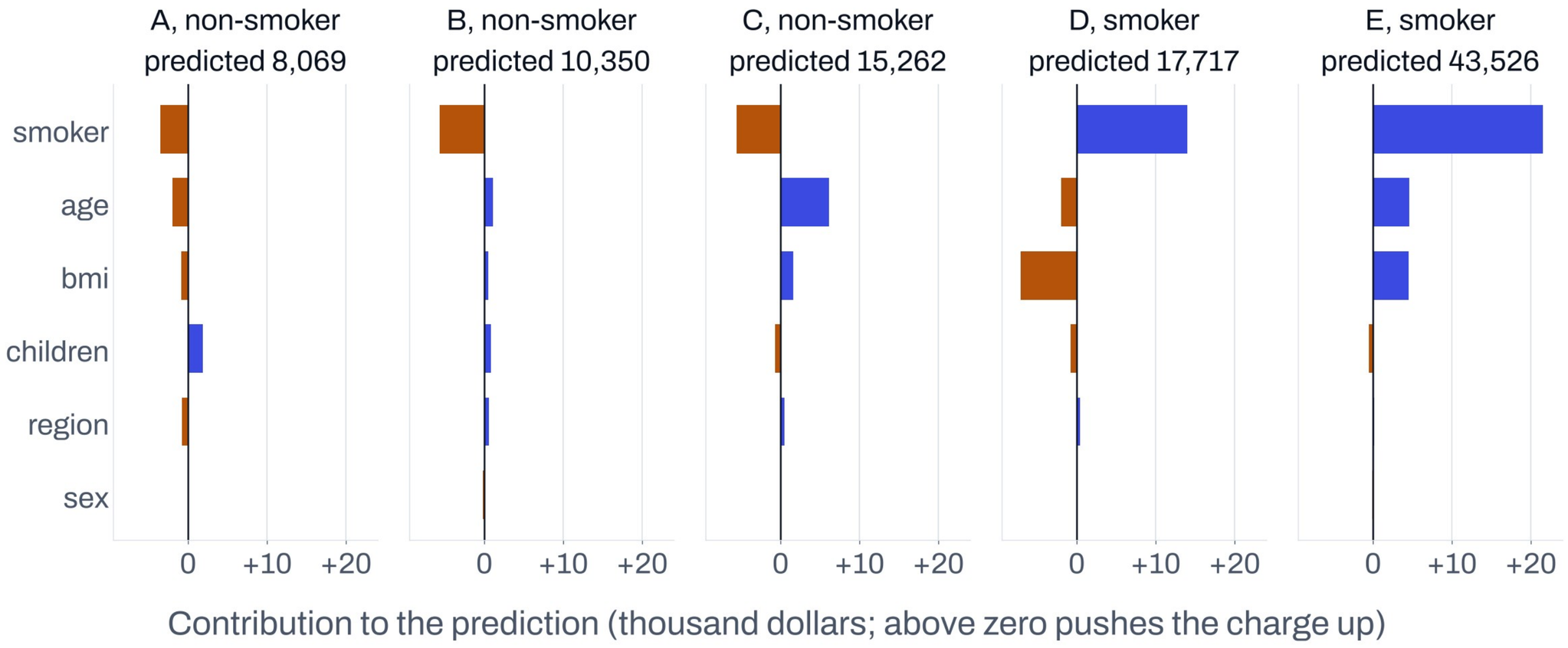
```
SELECT charges, ROUND(predicted_charges) AS prediction,
       ROUND(baseline_prediction_value) AS baseline,
       top_feature_attributions
FROM ML.EXPLAIN_PREDICT(MODEL ml.ins_boosted,
  (SELECT age, sex, bmi, children, smoker, region, charges
   FROM ml.insurance_groups
   WHERE split = 'validation'
     AND ROUND(charges, 2) IN (4391.65, 10370.91, 13228.85, 17904.53, 43578.94)),
  STRUCT(6 AS top_k_features))
ORDER BY charges;
```

→ baseline about 13,219 · E: smoker about +21,546

Trees take 10–15 minutes or more to train. Not ready? labs/D15/Resources/fallback/.

FIVE VALIDATION PEOPLE · ONE BAR PER FEATURE

Smoking pushes D and E up, and A, B and C down



Indigo pushes the predicted charge up, amber pulls it down. Thousands of dollars a year.

INSTRUCTOR DO

Step 5.3

The model overall

Average the size of every push over many people, then compare with D14's feature importance.

Suggested time · 8 minutes

STEP 3 · THE GLOBAL EXPLANATION

Switch it on when the model is trained

enable_global_explain = TRUE: ML.GLOBAL_EXPLAIN needs it at training.

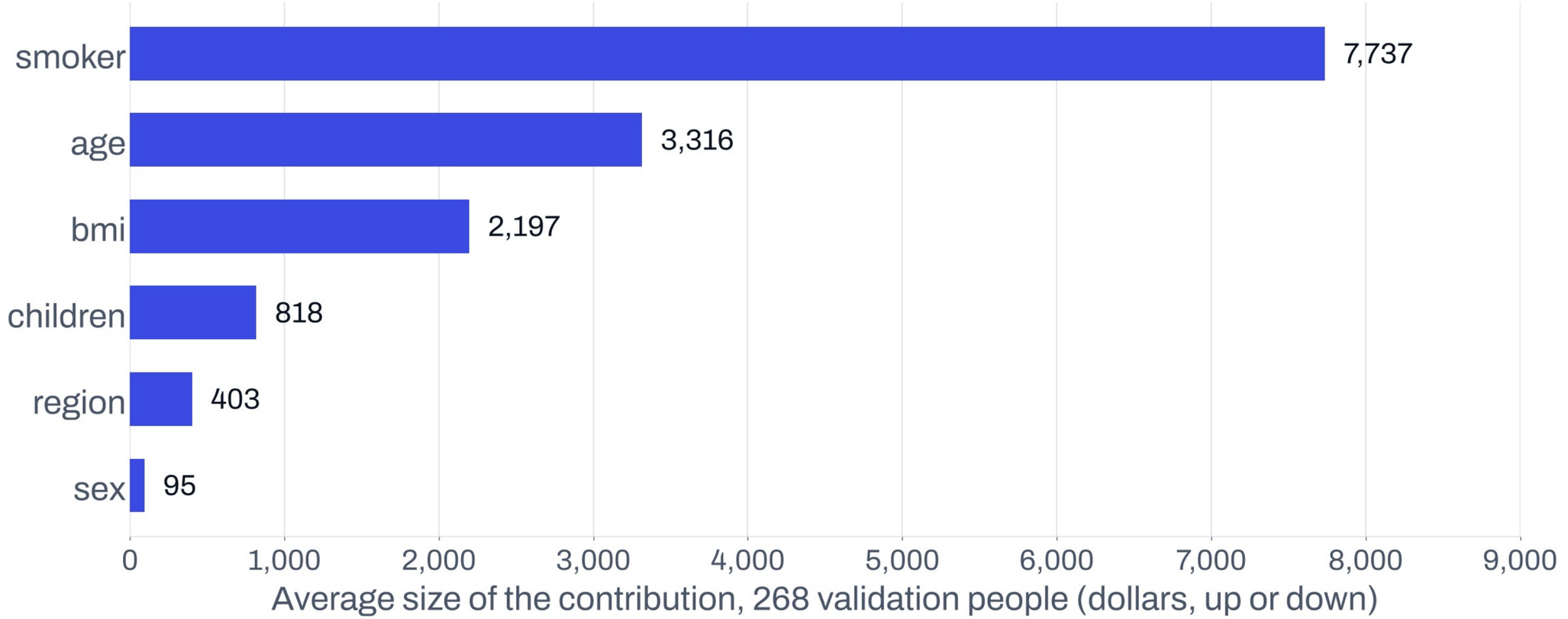
```
CREATE OR REPLACE MODEL ml.ins_explain
OPTIONS (model_type = 'BOOSTED_TREE_REGRESSOR',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT',
         enable_global_explain = TRUE) AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups WHERE split = 'train';
SELECT feature, ROUND(attribution) AS attribution
FROM ML.GLOBAL_EXPLAIN(MODEL ml.ins_explain)
ORDER BY attribution DESC;

                → smoker about 7,737 · age about 3,316 · bmi about 2,197
```

A stand-in over the validation people: compare the order.

A STAND-IN FOR ML.GLOBAL_EXPLAIN · AVERAGE SIZE OF EACH PUSH · DOLLARS

Smoking moves a typical prediction most



D14’s ML.FEATURE_IMPORTANCE: smoker about 0.96 of the gain. Both describe the model, not what causes charges.

STUDENT DO

Explain person D to the pricing team

One sentence on why D’s predicted charge is what it is, using his pushes from ML.EXPLAIN_PREDICT.

Suggested time

7 minutes

labs/D15/07-Evr-Explain-Five-People/README.md

STUDENT DO

One sentence for the pricing team

INSTRUCTIONS	Person D: 33, male, smoker, BMI 24.8, predicted about 17,717 dollars. Write one sentence a pricing manager can read: what pushed his charge up, what pulled it down, and by how much.
HINT	D is a smoker with a low BMI. Look at his BMI bar.
BONUS	Does BMI push the same way for person E? Why not?

Full steps and Check yourself: labs/D15/07-Evr-Explain-Five-People/README.md

A SAMPLE ANSWER · PERSON D · BASELINE ABOUT 13,219

Smoking adds; a low BMI takes some back

Feature	Push, dollars (about)
smoker: yes	**+13,987**
bmi 24.8	**−7,144**
age 33	−1,959
children, region, sex	−386
prediction	17,717

“Smoking adds about 14,000 dollars to D’s charge; his BMI of about 25 takes about 7,100 back.” E’s higher BMI, 31.2, adds about 4,553 to hers.

PART 05

Fix the leak, then serve

Cut points from the training rows only, then two tables Power BI can read.

INSTRUCTOR DO

Step 6.1

TRANSFORM keeps the training rules

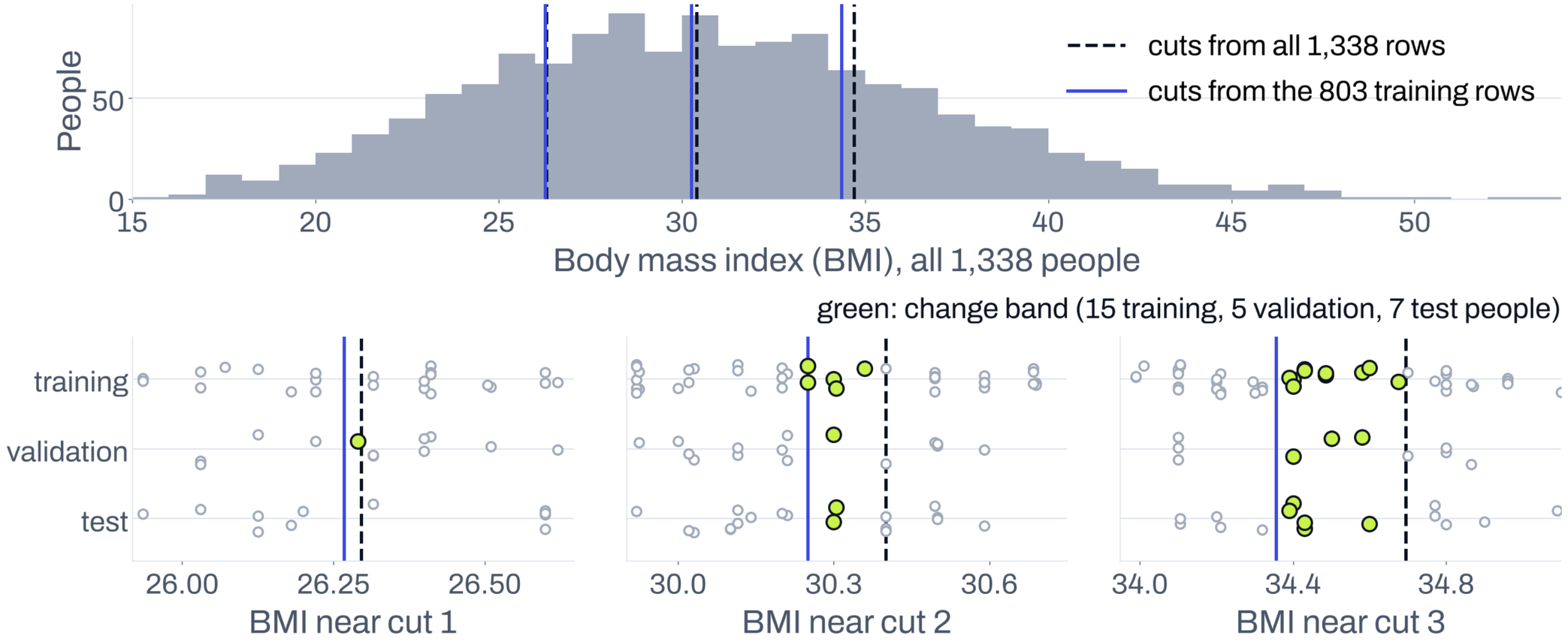
D14 lab 10 cut BMI into four bands using every person, test people included.

Suggested time · 7 minutes

labs/D15/08-Evr-Transform-And-Save/README.md

BMI QUARTILE CUTS · ALL 1,338 PEOPLE AGAINST THE 803 TRAINING PEOPLE

The test people helped choose the cut points



All rows: 26.30 · 30.40 · 34.69. Training only: 26.27 · 30.25 · 34.36. 5 validation, 7 test people move.

DEFINITION

TRANSFORM puts the preprocessing inside the model: the cut points are computed from the training rows and reused, unchanged, at every prediction.

STEP 1 · PREPROCESSING INSIDE THE MODEL

Four BMI bands, cut on the training rows

TRANSFORM sees only the rows the SELECT gives it: the 803 training people.

```
CREATE OR REPLACE MODEL ml.ins_transform
TRANSFORM (ML.QUANTILE_BUCKETIZE(bmi, 4) OVER () AS bmi_band,
           age, sex, children, smoker, region, charges)
OPTIONS (model_type = 'LINEAR_REG',
         input_label_cols = ['charges'],
         data_split_method = 'NO_SPLIT') AS
SELECT age, sex, bmi, children, smoker, region, charges
FROM ml.insurance_groups
WHERE split = 'train';
```

Google: “the same statistics are automatically applied to the input in prediction.”

EVERYONE DO

Step 6.2

Save predictions where Power BI can read them

Two ml.* tables: the 1960 forecast and every insured person scored.

Suggested time · 8 minutes

labs/D15/08-Evr-Transform-And-Save/README.md

STEP 2 · A TABLE FOR THE REPORT

The 1960 forecast, saved as a table

CREATE OR REPLACE TABLE: the sandbox runs no INSERT, so one statement writes the table.

```
CREATE OR REPLACE TABLE ml.air_forecast AS
SELECT DATE(f.forecast_timestamp) AS month,
       ROUND(f.forecast_value, 1) AS forecast_passengers,
       ROUND(f.prediction_interval_lower_bound, 1) AS lower_95,
       ROUND(f.prediction_interval_upper_bound, 1) AS upper_95,
       te.passengers AS actual_passengers
FROM ML.FORECAST(MODEL ml.air_arima_final,
                 STRUCT(12 AS horizon, 0.95 AS confidence_level)) AS f
LEFT JOIN ml.air_test AS te ON te.month = DATE(f.forecast_timestamp);

                                  → 12 rows: 1960, forecast beside actual
```

Then ml.insurance_scored, 1,338 rows; no table shares a column’s name.

BigQuery ML functions today

In class	What it gives you	Activity
ML.FORECAST	the next months, each with a lower and upper bound	02, 04–06, 08
ML.EXPLAIN_FORECAST	each month split into trend, season, spikes and dips	04
ML.EXPLAIN_PREDICT	each prediction with the features that pushed it	07
ML.GLOBAL_EXPLAIN	how much each feature drives predictions, on average	07
ML.FEATURE_IMPORTANCE	trees only: how often and how usefully each feature split	07
ML.QUANTILE_BUCKETIZE	bands of about equal size; TRANSFORM keeps the cut points	08
ML.PREDICT	a prediction for each row you pass in	08

Lab	New in the labs
09, 10, 13	**ML.ARIMA_EVALUATE · ML.ARIMA_COEFFICIENTS**
10	**ML.DETECT_ANOMALIES (billed, even in a sandbox) · ML.EVALUATE · ML.METRICS (Preview)**
12	**ML.TRANSFORM · ML.FEATURE_INFO · ML.BUCKETIZE · ML.FEATURE_CROSS · ML.POLYNOMIAL_EXPAND**
12	**ML.STANDARD_SCALER · ML.MIN_MAX_SCALER · ML.MAX_ABS_SCALER · ML.ROBUST_SCALER**
12	**ML.ONE_HOT_ENCODER · ML.LABEL_ENCODER · ML.IMPUTER · more in the lab’s table**
14	**ML.GET_INSIGHTS, on a CONTRIBUTION_ANALYSIS model**

CLOSE

What you can do now

- Forecast a series; read the interval
- Split by date; beat simple forecasts
- Explain a prediction and the model
- Fix a leak with TRANSFORM; save ml.* tables

NEXT · D16 The capstone on new public data, then Power BI.

AFTER CLASS · SELF-GUIDED · NOT SUBMITTED

Labs 09 to 14, option by option

Lab	Folder in labs/D15/	What you try
09	09-Lab-Arima-Plus-Options	every CREATE MODEL option, one per step; checkpoints C1–C3; the log fix
10	10-Lab-Forecast-Functions	ML.FORECAST, ML.EXPLAIN_FORECAST, ML.ARIMA_EVALUATE; ML.DETECT_ANOMALIES, billed even in the sandbox
11	11-Lab-Explain-Options	ML.EXPLAIN_PREDICT and ML.GLOBAL_EXPLAIN options, beside ML.FEATURE_IMPORTANCE
12	12-Lab-Transform-Functions	TRANSFORM with bucketing and scaling; ML.TRANSFORM shows the features
13	13-Lab-Nakheel-Daily-Revenue	optional: Nakheel’s daily revenue, forecast and saved for D22
14	14-Lab-Nakheel-Revenue-Drivers	optional: which slices explain Nakheel’s revenue change (ML.GET_INSIGHTS)